Step 2b - H6 - Millers enzyme inactivation without steam

Tech baseline

Technologies used in the market

Direct-steam hydrothermal treatment (status quo)

How it works: Saturated steam (100-120 C, ~1-3 bar) injected onto bran/germ/grain in a screw conveyor, kiln or vibratory tower, followed by re-drying and cooling. Sold as kilns (Buehler Granotherm, Streckel & Schrader) or continuous stabilizers (Revtech).

How well it performs: >95% lipase inactivation with atmospheric steaming, complete peroxidase inactivation; grain steaming 180 s inactivates lipase/peroxidase/xylanase without starch gelatinization; gives 6-18 months shelf life (Glanbia cites 18 months on oat flour).

Limits: Requires boiler, water treatment, steam distribution, condensate handling AND a re-drying step - high capex and floor space; uneconomic for small/medium mills; this is exactly the infrastructure the hypothesis removes.

Dry extrusion / expander cooking

How it works: Single-screw extruder uses friction + pressure to hit 120-140 C in seconds, no boiler; dominant method for rice bran stabilization in Asia and oilseeds.

How well it performs: Best-in-class enzyme kill: residual lipase and LOX <5% of fresh wheat germ in a 2023 LWT comparison; rice-bran FFA kept low for months; ~400 kg/h low-cost units, specific energy ~0.06-0.07 kWh/kg.

Limits: Severe mechanical treatment changes texture/functionality (fine for feed/oil bran, problematic for premium food-grade germ or functional flour); high screw wear; high shear not always wanted in a flour line.

Dry heat: fluid bed, oven/roasting, infrared

How it works: Hot air or IR heats product at 130-160 C for minutes; vibrated fluidized bed (VFB) is the gentlest variant.

How well it performs: 160 C oven heating ranks second to steaming for LOX inactivation; VFB gave the best lipid stability and tocopherol retention in comparative studies.

Limits: Dry heat transfers poorly into particles - longer residence times and less complete lipase inactivation than moist heat; still needs a heat source and doesn't remove the 'why not just use steam' question unless electrified.

Steam explosion (batch)

How it works: Bran rehydrated, pressurized at 0.8 MPa / 170 C for ~5 min, then explosively depressurized.

How well it performs: Inactivates lipase and peroxidase fully; improves soluble fibre and antioxidant extraction vs. autoclaving.

Limits: Batch, high-pressure vessel, needs boiler anyway - opposite of the retrofit-electric value proposition.

Continuous microwave (the hypothesis)

How it works: 2.45 GHz volumetric dielectric heating raises product to 60-100+ C in minutes while simultaneously drying (lower water activity adds microbial and enzymatic stability).

Lab/pilot studies on wheat germ and bran exist; continuous tunnel equipment is generic (SAIREM, Pueschner, Chinese OEMs).

How well it performs: 3-4 kW, ~8 min continuous microwave cut wheat-germ residual lipase to ~14-17% and fully inactivated LOX, with good flavor/color; 60-70 C microwave for 5-20 min inactivated lipase with no color/functionality change; combined steam+microwave achieves complete inactivation.

Limits: Lipase is harder to kill than LOX - microwave alone may stall at ~85% inactivation vs. steam's >95%; low-moisture bran is nearly transparent to microwaves (dielectric constant <1.9) so heating efficiency drops as it dries; uniformity at tonne/h scale and ATEX compliance are unproven industrially.

Ohmic heating (emerging)

How it works: Electric current passed through moistened bran heats it volumetrically.

How well it performs: 10-kg batch kept rice-bran FFA at 4.8% after 75 days vs 41.8% untreated.

Limits: Research stage, needs ~30-40% moisture (re-wetting + re-drying), electrodes contact issues - not industrialized.

Why our solution can be better

An all-electric microwave inactivation unit removes the boiler, steam distribution, water treatment and separate re-dryer that make steam stabilization uneconomic for the thousands of SME mills - it can slot into an existing bran/germ stream as a compact skid with instant on/off and no combustion permits. Microwave heats volumetrically and dries while it inactivates, so the product leaves shelf-stable in one pass, and electrification aligns with EU decarbonization incentives. The honest catch: microwave alone is proven only at lab/pilot scale, lipase is thermally robust, and dry bran absorbs microwaves poorly - a light steam-assist or pre-humidification may still be needed, partially re-importing the complexity being removed.

Key assumptions

• Global flour output ~285-450 Mt/yr; whole-grain and stabilized bran/germ streams addressable maybe 5-15% of that (assumed)

• Bran is ~20-25% and germ ~2-3% of wheat mass at milling; stabilization applies mainly to these fractions plus oat groats and rice bran (measured)

• Steam stabilization line capex (boiler + kiln + dryer) is in the high six to seven figures EUR and only pays at large-mill throughputs >1-2 t/h (assumed)

• Microwave wall-to-product efficiency ~50-60%; industrial electricity ~0.10-0.15 EUR/kWh vs gas ~0.03-0.05 EUR/kWh in EU - OPEX parity needs higher energy efficiency or smaller treated mass (assumed)

• Target spec: >=90% lipase inactivation and full LOX/peroxidase kill at 0.5-2 t/h continuous, product exit moisture <=12%, ATEX Zone 21/22-compliant execution (assumed)

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